

Training: linear (ridge)

Closed-form ridge regression on the lagged dRR.

A global ridge regression, solved in closed form in 0.6 s. No epochs and no int8 export (10 parameters).

How it is trained

- Inputs: the 'ar' feature set, rr_dev, drr_lag0, drr_lag1, drr_lag2, drr_lag3, drr_lag4, drr_lag5, drr_lag6, drr_lag7, z-scored with training statistics.
- Target: dRR[n+1] z-scored, NOT residual (the model is a global AR, the counterpart of the per-patient rls_ar).
- Solver: $A = [\text{features}, 1]$ for all 20,755,925 training windows; $A'A$ and $A'z$ are accumulated in float64 in chunks of 1M windows (no 20M x 10 float64 copy), then $w = \text{solve}(A'A + 1e-3 I, A'z)$ with the intercept left unpenalised. Exact least squares in one pass, so no learning rate, epochs or early stopping.
- Command: `python src/train.py --models linear`. Training time 0.6 s on CPU.

Fitted weights (z units)

input	weight
rr_dev	-0.2381
drr_lag0	-0.2877
drr_lag1	-0.2988
drr_lag2	-0.1263
drr_lag3	-0.0023
drr_lag4	-0.0055
drr_lag5	-0.0149
drr_lag6	-0.0254
drr_lag7	-0.0206
intercept	+0.0000

Outcome

- Validation RMSE 40.46 ms vs rls_ar 37.14: worse than the adaptive baseline overall and in every age interval (up to +24.5%), so rejected. A single set of weights for all patients cannot follow per-patient dynamics.

Abbreviations and acronyms used in this document

term	meaning
AR	autoregressive: a value predicted as a weighted sum of past values; AR(p) uses p past values
ar	feature-set name: autoregressive inputs (rr_dev and the 8 most recent dRR) of the linear model
CPU	central processing unit
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
drr_lag0..7	feature: drr_lag0..7: the 8 most recent dRR (lag0 = newest)
float64	64-bit (double-precision) floating-point numbers
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
ms	milliseconds
rls_ar	baseline: adaptive per-patient AR(8) forecast by RLS
RMSE	root mean square error (ms): the main accuracy measure, lower is better
rr_dev	feature: newest RR minus the window mean
z	z-score: $(\text{value} - \text{mean}) / \text{standard deviation}$, with statistics from the training subjects